Cover Page for Proposal Submitted to the National Aeronautics and Space Administration	NASA Proposal Number TBD on Submit

NASA PROCEDURE FOR HANDLING PROPOSALS

This proposal shall be used and disclosed for evaluation purposes only, and a copy of this Government notice shall be applied to any reproduction or abstract thereof. Any authorized restrictive notices that the submitter places on this proposal shall also be strictly complied with. Disclosure of this proposal for any reason outside the Government evaluation purposes shall be made only to the extent authorized by the Government.

SECTION I - Proposal Information

Principal Investigator	E-mail Address	Phone Number
Takashi Sato		

Street Address (1)	Street Address (2)

City	State / Province	Postal Code	Country Code

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

Proposed Start Date	Proposed End Date	Total Budget	Year 1 Budget	Year 2 Budget
03 / 02 / 2026	02 / 23 / 2030	0.00	0.00	0.00

SECTION II - Application Information

NASA Program Announcement Number	NASA Program Announcement Title
NNH25ZDA001N-COMPASS	F.19 Collaborative Opportunities for Mentorship, Partnership and Academic Success in Science

For Consideration By NASA Organization *(the soliciting organization, or the organization to which an unsolicited proposal is submitted)*

NASA , Headquarters , Science Mission Directorate , Cross Division

Date Submitted	Submission Method	Grants.gov Application Identifier	Applicant Proposal Identifier
	Electronic Submission Only		XP-9924-FIS

Type of Application	Predecessor Award Number	Other Federal Agencies to Which Proposal Has Been Submitted
New		NASA Glenn Research Center

International Participation	Type of International Participation
Yes	PI CO-I Equipment Facility Collaborator

SECTION III - Submitting Organization Information

UEI	EFT	CAGE Code	Employer Identification Number (EIN or TIN)	Organization Type

Organization Name (Standard/Legal Name)	Company Division
Organization DBA Name	Division Number

Street Address (1)	Street Address (2)

City	State / Province	Postal Code	Country Code

SECTION IV - Proposal Point of Contact Information

Name	Email Address	Phone Number
Takashi Sato	shakatheshaka@gmail.com	999969676 x 11

SECTION V - Certification and Authorization

Certification of Compliance with Applicable Executive Orders and U.S. Code

By submitting the proposal identified in the Cover Sheet/Proposal Summary in response to this Research Announcement, the Authorizing Official of the proposing organization (or the individual proposer if there is no proposing organization) as identified below:

- certifies that the statements made in this proposal are true and complete to the best of their knowledge;
- agrees to accept the obligations to comply with NASA award terms and conditions if an award is made as a result of this proposal; and
- confirms compliance with all provisions, rules, and stipulations set forth in this solicitation.

Willful provision of false information in this proposal and/or its supporting documents, or in reports required under an ensuing award, is a criminal offense (U.S. Code, Title 18, Section 1001).

Authorized Organizational Representative (AOR) Name	AOR E-mail Address	Phone Number

AOR Signature *(Must have AOR's original signature. Do not sign "for" AOR.)* Date

Digitally signed by

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION VI - Team Members

Team Member Role	Team Member Name	Contact Phone	E-mail Address	
PI	**Takashi Sato**			
Organization/Business Relationship		UEI	EFT	CAGE Code
International Participation	U.S. Government Agency		Total Funds Requested	
No			**0.00**	

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION VII - Project Summary

Project Title: ARTEMIS-CORE: Scalable Micro-Fission Reactors for Lunar Surface Power and Deep Space Nuclear Thermal Propulsion.

Summary:
The ARTEMIS-CORE project addresses the critical energy infrastructure gap for NASA's long-duration missions. As solar power is inherently limited by the 354-hour lunar night and the high-opacity dust environment of Mars, this proposal introduces a robust 40 kWe HALEU-fueled (High-Assay Low-Enriched Uranium) micro-fission reactor.

Technical Approach: The system utilizes a fast-spectrum core designed for maximum mass-to-power density, regulated by redundant Beryllium Oxide (BeO) sliding reflectors. Thermal energy is extracted via a primary NaK (Sodium-Potassium) liquid-metal loop, ensuring high-temperature stability without mechanical pump reliance. Power conversion is achieved through a high-efficiency Closed Brayton Cycle (CBC), targeting a system specific mass of <15 kg/kWe.

Mission Impact: ARTEMIS-CORE is dual-purpose:

Fission Surface Power (FSP): Sustaining In-Situ Resource Utilization (ISRU) plants and life-support systems for the Artemis Base Camp.

Nuclear Thermal Propulsion (NTP): Providing an Isp of ~900s for crewed Mars transit via Advanced Hydrogen Fission Propulsion.

Planetary Protection & Safety: Advanced shadow shielding (Tungsten/LiH) maintains crew dosage within ALARA limits (<5 rem/yr). Post-mission decommissioning strategy ensures localized activation of regolith within a 5m radius, adhering to COSPAR planetary protection guidelines.

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION VIII - Other Project Information

Proprietary Information

Is proprietary/privileged information included in this application?

Yes

International Collaboration

Does this project involve activities outside the U.S. or partnership with International Collaborators?

Yes

Principal Investigator	Co-Investigator	Collaborator	Equipment	Facilities
Yes	**Yes**	**Yes**	**Yes**	**Yes**

Explanation :

The system utilizes a fast-spectrum core designed for maximum mass-to-power density, regulated by redundant Beryllium Oxide (BeO) sliding reflectors. Thermal energy is extracted via a primary NaK (Sodium-Potassium) liquid-metal loop, ensuring high-temperature stability without mechanical pump reliance. Power conversion is achieved through a high-efficiency Closed Brayton Cycle (CBC), targeting a system specific mass of <15 kg/kWe.

NASA Civil Servant Project Personnel

Are NASA civil servant personnel participating as team members on this project (include funded and unfunded)?

Yes

Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year
2026	**2028**	**2029**			
Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs
2026.01	**2028.10**	**2029.08**			

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION VIII - Other Project Information

Environmental Impact

Does this project have an actual or potential impact on the environment?	Has an exemption been authorized or an environmental assessment (EA) or an environmental impact statement (EIS) been performed?
Yes	**Yes**

Environmental Impact Explanation:

Terrestrial Testing: Controlled radiological risks during ground-based assembly tests. Launch Safety: Potential for radiological release in event of vehicle failure; mitigated by "cold" launch state (non-critical during ascent). Planetary Activation: Localized activation of regolith within a 5m radius.

Exemption/EA/EIS Explanation:

A preliminary Environmental Assessment (EA) has been performed in alignment with the NEPA and NASA's "Final Programmatic Environmental Impact Statement for Fission Surface Power Systems" (2023).

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION VIII - Other Project Information

Historical Site/Object Impact

Does this project have the potential to affect historic, archeological, or traditional cultural sites (such as Native American burial or ceremonial grounds) or historic objects (such as an historic aircraft or spacecraft)?

No

Explanation:

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION IX - Program Specific Data

Question 1 : Short Title:
Answer: AOI (Cosmic Vigilant)

Question 2 : SMD Division
Answer: Astrophysics

Question 3 : Cross-Divisional
Answer: Yes

Question 4 : Multiple Division selection
Answers:
- **Astrophysics**
- **Biological & Physical Sciences**
- **Earth Science**
- **Heliophysics**
- **Planetary Science**

Question 5 : Number of students
Answer: 1500

Question 6 : Student support
Answer: 30%

Question 7 : COMPASS Eligible Institution support
Answer: 30%

Question 8 : Length of Award in years
Answer: 2

Question 9 : NASA Center/Facility collaboration
Answer: Jet Propulsion Laboratory

Question 10 : Carnegie Classification
Answer: Very highly research intensive (R1)

Question 11 : Type of institution:
Answer: NASA Center (including JPL)

Question 12 : Will any funding be provided to a federal government organization including NASA Centers, JPL, other Federal agencies, government laboratories, or Federally Funded Research and Development Centers (FFRDCs)?
Answer: Yes

Question 13 : Is this Federal government organization a different organization from the proposing (PI) organization?
Answer: Yes

Question 14 : Does this proposal include the use of NASA-provided high end computing (HEC)?
Answer: Yes

Question 15 : HEC Request Number
Answer: HEC-SMD-26-9924

Question 16 : Research Category:
Answer: 1) Theory/computer modeling

Question 17 : Flight Services
Answer: No

Question 18 : Team members not confirmed via NSPIRES
Answer:
Anthony Calomino (Space Nuclear); Jason Derleth (NIAC)

Question 19 : Does this proposal contain information and/or data that are subject to U.S. export control laws and regulations including Export Administration Regulations (EAR) and International Traffic in Arms Regulations (ITAR)?

Answer: No

Question 20 : I have identified the export-controlled material in this proposal.
Answer: No

Question 21 : I acknowledge that the inclusion of such material in this proposal may complicate the government's ability to evaluate the proposal.
Answer: No

Question 22 : Does the proposed work include any involvement with collaborators in China or with Chinese organizations, or does the proposed work include activities in China?
Answer: No

The National Environmental Policy Act (NEPA) obligates NASA to consider the potential environmental effects of proposed projects, including those that NASA funds which are implemented by grantees. The majority of grant-related activities are categorically excluded as research and development projects that do not pose adverse environmental impacts. The following questions enable NASA to ascertain whether your proposal will require additional NEPA analysis if selected (e.g., filling out an Environmental Checklist) or the completion of NASA's Executive Order (EO) 12114 Checklist for an activity to be conducted abroad. "Yes" responses are not selection criteria, however, if a "Yes" response is marked, proposers should consider NEPA and/or EO compliance in cost and schedule estimates.

Question 23 : Would the proposal involve any activity that includes: a. Construction of new facilities or modification to the footprint of an existing-facility, or b. Ground disturbance (e.g., excavation, clearing of trees, installation of equipment, etc.), or c. Outdoor discharges of water (e.g., waste water runoff), air emissions (e.g., ozone-depleting substances) or generation of noise exceeding 115 dBA (excluding those associated with aircraft operations)?
Answer: No

Question 24 : Would the proposal involve any field activity that would: a. Release equipment (e.g., dropsondes, sensors, etc.) or chemicals (e.g., dyes, tracers, etc.) into the air, bodies of water or on the ground, or b. Release a parachute or use equipment that would not be recovered, or c. Involve equipment or a payload that contains hazardous (e.g., petroleum, hypergols, oxidizers,solid propellants, etc.) or radioactive materials?
Answer: No

Question 25 : Would the proposal involve the launch of a payload, equipment, or instrument (e.g., via launch vehicle, sounding rocket, balloon, etc.)?
Answer: No

Question 26 : Would the proposal involve any activity to be conducted outside the United States or its territories excluding travel for meetings or conferences?
Answer: Yes

Question 27 : Comments
Answer:
(AOI Cosmic Vigilant for autonomous reactivity control)

Question 28 : Does this proposal contain an AI or ML component?
Answer: Yes

Question 29 : Primary Technology
Answer: TX11: Software, Modeling, Simulation, and Information Processing

Question 30 : Agency Priorities
Answers:
NASA's Moon to Mars Strategy enabling U.S. astronauts to return to the Moon and prepare for Mars exploration;
The study of space weather and/or asteroids to protect American security and economic interests
The utilization of Artificial Intelligence, Machine Learning and Quantum Information Science to drive innovation
Innovative analyses of existing data that maximize the value of previous NASA SMD investments in space science data collection
Earth Science remote sensing that delivers tangible benefits to American society and the economy

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION X - Budget

Cumulative Budget

Budget Cost Category	Funds Requested ($)		
	Year 1 ($)	Year 2 ($)	Total Project ($)
A. Direct Labor - Key Personnel	0.00	0.00	0.00
B. Direct Labor - Other Personnel	0.00	0.00	0.00
Total Number Other Personnel	0	0	0
Total Direct Labor Costs (A+B)	0.00	0.00	0.00
C. Direct Costs - Equipment	0.00	0.00	0.00
D. Direct Costs - Travel	0.00	0.00	0.00
Domestic Travel	0.00	0.00	0.00
Foreign Travel	0.00	0.00	0.00
E. Direct Costs - Participant/Trainee Support Costs	0.00	0.00	0.00
Tuition/Fees/Health Insurance	0.00	0.00	0.00
Stipends	0.00	0.00	0.00
Travel	0.00	0.00	0.00
Subsistence	0.00	0.00	0.00
Other	0.00	0.00	0.00
Number of Participants/Trainees			0
F. Other Direct Costs	0.00	0.00	0.00
Materials and Supplies	0.00	0.00	0.00
Publication Costs	0.00	0.00	0.00
Consultant Services	0.00	0.00	0.00
ADP/Computer Services	0.00	0.00	0.00
Subawards/Consortium/Contractual Costs	0.00	0.00	0.00
Equipment or Facility Rental/User Fees	0.00	0.00	0.00
Alterations and Renovations	0.00	0.00	0.00
Other	0.00	0.00	0.00
G. Total Direct Costs (A+B+C+D+E+F)	0.00	0.00	0.00
H. Indirect Costs	0.00	0.00	0.00
I. Total Direct and Indirect Costs (G+H)	0.00	0.00	0.00
J. Fee	0.00	0.00	0.00
K. Total Cost (I+J)	0.00	0.00	0.00
Total Cumulative Budget			0.00

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION X - Budget

Start Date :	End Date :	Budget Type : **Project**	Budget Period : **1**

A. Direct Labor - Key Personnel

Name	Project Role	**Base Salary ($)**	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	**Fringe Benefits ($)**	Funds Requested ($)
Sato, Takashi	**PI**	**0.00**				**0.00**	**0.00**	**0.00**
							Total Key Personnel Costs	**0.00**

B. Direct Labor - Other Personnel

Number of Personnel	Project Role	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
0	Total Number Other Personnel					Total Other Personnel Costs	**0.00**
						Total Direct Labor Costs (Salary, Wages, Fringe Benefits) (A+B)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

C. Direct Costs - Equipment

Item No.	Equipment Item Description	Funds Requested ($)
	Total Equipment Costs	**0.00**

D. Direct Costs - Travel

	Funds Requested ($)
1. Domestic Travel (Including U.S. Territories and Possessions)	**0.00**
2. Foreign Travel (Including Canada and Mexico)	**0.00**
Total Travel Costs	**0.00**

E. Direct Costs - Participant/Trainee Support Costs

		Funds Requested ($)
1. Tuition/Fees/Health Insurance		**0.00**
2. Stipends		**0.00**
3. Travel		**0.00**
4. Subsistence		**0.00**
Number of Participants/Trainees:	Total Participant/Trainee Support Costs	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

F. Other Direct Costs

	Funds Requested ($)
1. Materials and Supplies	**0.00**
2. Publication Costs	**0.00**
3. Consultant Services	**0.00**
4. ADP/Computer Services	**0.00**
5. Subawards/Consortium/Contractual Costs	**0.00**
6. Equipment or Facility Rental/User Fees	**0.00**
7. Alterations and Renovations	**0.00**
8. Other:	**0.00**
9. Other:	**0.00**
10. Other:	**0.00**
11. Other:	**0.00**
12. Other:	**0.00**
13. Other:	**0.00**
14. Other:	**0.00**
15. Other:	**0.00**
16. Other:	**0.00**
17. Other:	**0.00**
Total Other Direct Costs	**0.00**

G. Total Direct Costs

	Funds Requested ($)
Total Direct Costs (A+B+C+D+E+F)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

H. Indirect Costs

	Indirect Cost Rate (%)	**Indirect Cost Base ($)**	Funds Requested ($)
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
Cognizant Federal Agency:		Total Indirect Costs	**0.00**

I. Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Costs (G+H)	**0.00**

J. Fee

	Funds Requested ($)
Fee	**0.00**

K. Total Cost

	Funds Requested ($)
Total Cost with Fee (I+J)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION X - Budget

Start Date :	End Date :	Budget Type : **Project**	Budget Period : **2**

A. Direct Labor - Key Personnel

Name	Project Role	**Base Salary ($)**	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	**Fringe Benefits ($)**	Funds Requested ($)
Sato, Takashi	**PI**	**0.00**				**0.00**	**0.00**	**0.00**
							Total Key Personnel Costs	**0.00**

B. Direct Labor - Other Personnel

Number of Personnel	Project Role	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
0	Total Number Other Personnel					Total Other Personnel Costs	**0.00**
						Total Direct Labor Costs (Salary, Wages, Fringe Benefits) (A+B)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

C. Direct Costs - Equipment

Item No.	Equipment Item Description	Funds Requested ($)
	Total Equipment Costs	**0.00**

D. Direct Costs - Travel

	Funds Requested ($)
1. Domestic Travel (Including U.S. Territories and Possessions)	**0.00**
2. Foreign Travel (Including Canada and Mexico)	**0.00**
Total Travel Costs	**0.00**

E. Direct Costs - Participant/Trainee Support Costs

		Funds Requested ($)
1. Tuition/Fees/Health Insurance		**0.00**
2. Stipends		**0.00**
3. Travel		**0.00**
4. Subsistence		**0.00**
Number of Participants/Trainees:	Total Participant/Trainee Support Costs	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

F. Other Direct Costs

	Funds Requested ($)
1. Materials and Supplies	**0.00**
2. Publication Costs	**0.00**
3. Consultant Services	**0.00**
4. ADP/Computer Services	**0.00**
5. Subawards/Consortium/Contractual Costs	**0.00**
6. Equipment or Facility Rental/User Fees	**0.00**
7. Alterations and Renovations	**0.00**
8. Other:	**0.00**
9. Other:	**0.00**
10. Other:	**0.00**
11. Other:	**0.00**
12. Other:	**0.00**
13. Other:	**0.00**
14. Other:	**0.00**
15. Other:	**0.00**
16. Other:	**0.00**
17. Other:	**0.00**
Total Other Direct Costs	**0.00**

G. Total Direct Costs

	Funds Requested ($)
Total Direct Costs (A+B+C+D+E+F)	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **ARTEMIS-CORE: Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship Proposal ID: XP-9924-FIS Lead Investigator: Sato (PI) A.I. Systems Lead: AOI (Cosmic Vigilant)**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**2**

H. Indirect Costs

	Indirect Cost Rate (%)	**Indirect Cost Base ($)**	Funds Requested ($)
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
Cognizant Federal Agency:		Total Indirect Costs	**0.00**

I. Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Costs (G+H)	**0.00**

J. Fee

	Funds Requested ($)
Fee	**0.00**

K. Total Cost

	Funds Requested ($)
Total Cost with Fee (I+J)	**0.00**

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

Space Technology Mission Directorate (STMD)

ARTEMIS-CORE

Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship

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Solicitation: F.19 Collaborative Opportunities for Mentorship, Partnership and Academic Success in Science (COMPASS25)
Proposal ID: XP-9924-FIS

Lead Investigator:	Sato (PI)
A.I. Systems Lead:	AOI (Cosmic Vigilant)
Lead Center:	NASA Glenn Research Center
Partners:	MSFC, Idaho National Laboratory

February 2026

1 Proposal Summary

Project Title: ARTEMIS-CORE: Scalable Micro-Fission Reactors for Lunar Surface Power and Deep Space Nuclear Thermal Propulsion.

1.1 Technical Abstract

The ARTEMIS-CORE project addresses the critical energy infrastructure gap for NASA's long-duration missions. As solar power is inherently limited by the 354-hour lunar night and the high-opacity dust environment of Mars, this proposal introduces a robust 40 kWe HALEU-fueled (High-Assay Low-Enriched Uranium) micro-fission reactor.

1.2 Technical Approach

The system utilizes a fast-spectrum core designed for maximum mass-to-power density, regulated by redundant Beryllium Oxide (BeO) sliding reflectors. Thermal energy is extracted via a primary NaK (Sodium-Potassium) liquid-metal loop, ensuring high-temperature stability without mechanical pump reliance. Power conversion is achieved through a high-efficiency Closed Brayton Cycle (CBC), targeting a system specific mass of < 15 kg/kWe.

1.3 Mission Impact

- **Fission Surface Power (FSP):** Sustaining In-Situ Resource Utilization (ISRU) plants and life-support systems for the Artemis Base Camp.
- **Nuclear Thermal Propulsion (NTP):** Providing an I_{sp} of ~ 900s for crewed Mars transit via Advanced Hydrogen Fission Propulsion.

2 Environmental and Supplemental Data

2.1 Environmental Impact Assessment (NEPA)

Impact Identified: Yes.
Explanation: Controlled radiological risks during ground testing and potential launch failure. Mitigation includes "cold launch" (non-critical ascent) and Cermet fuel containment.
Exemption/EA/EIS: Yes. Preliminary EA performed under the "Final Programmatic Environmental Impact Statement for Fission Surface Power Systems" (2023).

2.2 Cultural and Historic Site Impact

Impact Identified: No.
Explanation: All landing sites maintain a 2-km buffer from historic Apollo/Surveyor sites to prevent thermal/radiological interference.

3 NASA Civil Servant Personnel (FTEs)

Funding and personnel allocation for civil servants are requested in the following format: `0.X - XXXXX.XX`.

Fiscal Year	Period	FTE Allocation	Senior Role
2026	Year 1	0.5 - 2026.01	Senior Nuclear Engineer (NASA Glenn)
2027	Year 2	0.8 - 2027.05	Safety & Mission Assurance (NASA MSFC)
2028	Year 3	1.2 - 2028.10	Thermal Systems Integration (NASA JPL)
2029	Year 4	1.0 - 2029.08	Launch Integration Lead (NASA KSC)
2030	Year 5	0.4 - 2030.04	Post-Deployment Monitor (NASA JSC)

4 Administrative Metadata

- **Institution Type:** NASA Center (including JPL)
- **Research Category:** 6) Technology development applicable to space nuclear/electric propulsion
- **Export Control (ITAR/EAR):** Yes (Sections 3.2 and 5.1 restricted)
- **AI/ML Usage:** Yes (AOI - Cosmic Vigilant for autonomous reactivity control)
- **Relevant Divisions:** Planetary Science & Biological and Physical Sciences.
- **HEC Request:** HEC-SMD-26-9924

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

Space Technology Mission Directorate (STMD)

ARTEMIS-CORE

Collaborative Fission-Quantum Hybrid Systems for Deep Space Exploration and Academic Mentorship

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Solicitation: F.19 Collaborative Opportunities for Mentorship, Partnership and Academic Success in Science (COMPASS25)
Proposal ID: XP-9924-FIS

Lead Investigator:	Sato (PI)
A.I. Systems Lead:	AOI (Cosmic Vigilant)
Lead Center:	NASA Glenn Research Center
Partners:	MSFC, Idaho National Laboratory

February 2026

1 Proposal Summary

Project Title: ARTEMIS-CORE: Scalable Micro-Fission Reactors for Lunar Surface Power and Deep Space Nuclear Thermal Propulsion.

1.1 Technical Abstract

The ARTEMIS-CORE project addresses the critical energy infrastructure gap for NASA's long-duration missions. As solar power is inherently limited by the 354-hour lunar night and the high-opacity dust environment of Mars, this proposal introduces a robust 40 kWe HALEU-fueled micro-fission reactor.

1.2 Technical Approach

The system utilizes a fast-spectrum core designed for maximum mass-to-power density, regulated by redundant Beryllium Oxide (BeO) sliding reflectors. Thermal energy is extracted via a primary NaK liquid-metal loop. Power conversion is achieved through a high-efficiency Closed Brayton Cycle (CBC), targeting a system specific mass of < 15 kg/kWe.

2 Total Budget and Financial Plan

The total requested budget for Phase I (36 months) is **$4,750,000.00 USD**. This funding covers hardware prototyping, HALEU materials acquisition, and specialized labor.

Table 1: Total Budget Breakdown (USD)

Category	FY 2026	FY 2027	Total
Civil Servant Personnel (FTE)	$650,000	$720,000	$1,370,000
Contractor Labor (Engineering)	$850,000	$900,000	$1,750,000
Materials (HALEU & NaK)	$400,000	$250,000	$650,000
High End Computing (HEC) Fees	$120,000	$120,000	$240,000
Travel & Documentation	$40,000	$60,000	$100,000
Indirect Costs (Overhead)	$310,000	$330,000	$640,000
Grand Total	**$2,370,000**	**$2,380,000**	**$4,750,000**

3 NASA Civil Servant Personnel (FTEs)

Funding allocation for civil servants (FTE format: `0.X - XXXXX.XX`).

Fiscal Year	Period	FTE Allocation	Senior Role
2026	Year 1	0.5 - 2026.01	Senior Nuclear Engineer (NASA Glenn)
2027	Year 2	0.8 - 2027.05	Safety & Mission Assurance (NASA MSFC)
2028	Year 3	1.2 - 2028.10	Thermal Systems Integration (NASA JPL)

4 Environmental and Administrative Metadata

- **NEPA Status:** Preliminary EA performed (2023). Impact: Yes (Radiological).
- **HEC Number:** HEC-SMD-26-9924.
- **Export Control:** ITAR Restricted (Sections 3.2, 5.1).
- **Research Category:** 6) Technology development applicable to space nuclear propulsion.

Participant Information:

Participant Name: Armando Takashi Sato

Team Name: Taka System Inc co. ltda

Revolutionizing Digital Interaction with the AOI Platform

Page 1

Title: Neuromorphic Affective Flood Sentinel (NAFS): Intuitive, Rapid Flood Risk Assessment via Unconventional Computing

1. Introduction

A. Brief overview of proposed solution
This white paper proposes the Neuromorphic Affective Flood Sentinel (NAFS), an innovative solution leveraging neuromorphic computing to revolutionize rapid flood water analysis. NAFS aims to transform complex, multi-source flood data into an intuitive, quickly interpretable "affective state" assessment, providing emergency responders and decision-makers with immediate, actionable insights. Drawing inspiration from systems that synthesize diverse inputs into holistic understandings, NAFS will employ spiking neural networks (SNNs) on neuromorphic hardware to process fused data streams, dynamically assessing flood severity, trajectory, potential impact, and the confidence level of its own assessment. This approach moves beyond traditional numerical outputs to offer a more human-centric understanding of evolving flood events, enabling faster and more effective responses.

B. Description of team or organization
`[Your Team Name]` is a `[e.g., newly formed research group / small, agile development team / individual researcher]` passionate about applying cutting-edge computational paradigms to solve critical Earth science and societal challenges. Our expertise spans `[e.g., data simulation, front-end development for complex systems, and conceptual design of AI-driven interfaces – adapt from your AOI shell work]`.

We are driven by the potential of unconventional computing to unlock new capabilities in disaster management and environmental stewardship. This proposal represents our vision for a novel human-computer interaction model for complex event analysis, inspired by the AOI_draw_v4_shell's conceptual framework of an "affective" data processing entity.

C. Problem Statement & Audience

Current flood analysis methods, while increasingly sophisticated, often present challenges for rapid decision-making during emergencies. The sheer volume and variety of data from satellites (e.g., SAR, MODIS), weather models, hydrological simulations, and in-situ sensors can be overwhelming. Classical computational models may require significant processing time, and their outputs, typically detailed numerical maps or datasets, may not be immediately intuitive for on-the-ground emergency managers or policymakers under high-stress conditions.

Our **primary audience** is **emergency response coordinators and disaster management agencies** (e.g., FEMA, local government units). These users require:

1. **Speed:** Near real-time assessment of flood extent and severity.
2. **Clarity:** Easily understandable information that highlights critical changes and risks.
3. **Confidence:** An understanding of the reliability of the information presented.

NAFS addresses the need for a system that rapidly synthesizes diverse data into a high-level, "affective" summary of the flood situation, analogous to an intuitive grasp of a complex scenario.

D. Chosen Unconventional Computing Method

The core of NAFS will utilize **Neuromorphic Computing**, specifically leveraging Spiking Neural Networks (SNNs). This choice is driven by:

- **Efficiency:** SNNs on neuromorphic hardware are known for their low power consumption and high-speed processing of temporal, event-based data, ideal for real-time sensor streams.
- **Pattern Recognition:** SNNs excel at complex spatio-temporal pattern recognition and learning from sparse, noisy data, characteristic of evolving flood events.
- **Temporal Dynamics:** Their inherent ability to process information in the time domain is crucial for understanding the progression of floods.

We also envision a potential hybrid approach where **In-Memory Computing** could be utilized for the initial, high-throughput stages of data ingestion and pre-processing from diverse, large-scale datasets before feeding into the neuromorphic core, mitigating the von Neumann bottleneck for data-intensive fusion tasks.

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2. Conceptual Framework and System Architecture

The NAFS system is conceptualized as a multi-layered architecture designed to ingest, process, and interpret flood-related data, culminating in an "affective" risk assessment.

A. System Architecture Overview

(Conceptual Diagram 1: High-Level NAFS Architecture)

```
+---------------------------+     +-------------------------+     +----------
-------------+
|      Data Sources         | --> | Data Ingestion & Fusion | --> |
Neuromorphic Core   |
| (Satellite, Weather, IoT, |     | (Potentially In-Memory  |     | (SNN-
based Affective  |
|  Social Media, Models)    |     |  Computing Assisted)    |     |  State
Processing)    |
+---------------------------+     +-------------------------+     +----------
-------------+
                                                                          |
                                                                          |
                                                                          v
+-------------------------------------------------+     +---------------------
--------------+
| User Interface & Visualization                  | <-- | Affective State
Output & Confidence |
| (Dashboard, "AOI Shell" like query interface)  |     | (e.g., Risk Level,
Stability,     |
+-------------------------------------------------+     |  Urgency, System
Certainty)       |
                                                      +-----------------------
------------+
```

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B. Key Components

1. **Data Sources:**
    - **Satellite Imagery:** SAR (e.g., Sentinel-1), optical (e.g., MODIS, Landsat) for flood extent mapping.

- **Meteorological Data:** Precipitation forecasts (NWP models), radar data, real-time rainfall.
- **Hydrological Models:** Outputs from existing flood models (e.g., National Water Model).
- **In-situ Sensors:** River gauges, soil moisture sensors, weather stations.
- **Crowdsourced/Social Media Data:** Filtered reports and images indicating local impacts.
- **Geospatial Data:** Digital Elevation Models (DEMs), land use/land cover maps, infrastructure layers.

2. **Data Ingestion & Fusion Module:**
   - Responsible for collecting, pre-processing (normalization, geo-referencing), and fusing heterogeneous data streams.
   - **Challenge:** Handling varying data resolutions, latencies, and formats.
   - **Unconventional Opportunity:** In-Memory Computing could accelerate this stage by performing initial filtering, feature extraction, and co-registration of massive datasets directly where data resides, reducing data movement overhead.
   - Symbolically, let `D = {d_1, d_2, ..., d_n}` be the set of n raw data streams. The fusion process `F` yields a unified spatio-temporal representation `X(t) = F(D, t)`.
3. **Neuromorphic Core (The "Affective Sentinel"):**
   - This is the heart of NAFS, implemented with SNNs on neuromorphic hardware (e.g., Intel Loihi, IBM TrueNorth, or similar research platforms/simulators).
   - The fused data `X(t)` is encoded into spike trains. For example, changes in water extent, rainfall intensity, or sensor readings above thresholds would generate spikes.
   - The SNN architecture would consist of multiple layers/areas:
     - **Feature Extraction Layers:** Learning to identify relevant features from the spike patterns (e.g., rapid water level rise, intense localized rainfall over vulnerable areas).
     - **Associative Layers:** Learning correlations between these features and historical flood impact data or expert-defined rules.
     - **Affective State Generation Layer:** A higher-level network that integrates outputs from associative layers to produce a set of "affective" parameters.
   - **Learning:** The SNN would be trained using a combination of supervised learning (on historical flood events and their impacts), unsupervised learning (to discover novel patterns), and potentially reinforcement learning (based on feedback from human experts or simulated outcomes). Spike-Timing-Dependent Plasticity (STDP) or similar bio-inspired learning rules would govern synaptic weight adjustments.
4. **Affective State Output & Confidence Module:**
   - Translates the SNN's internal state into human-understandable "affective" indicators. These are not literal emotions but rather intuitive summaries. Examples:

        - **Overall Risk (OR):** Low, Moderate, High, Severe, Catastrophic.
        - **Dynamic Stability (DS):** Stable, Slowly Worsening, Rapidly Escalating, Stabilizing, Improving.
        - **Spatial Focus (SF):** Dispersed, Concentrating (Region X), Linear (River Y).
        - **Impact Potential (IP):** Infrastructure, Population, Agriculture.
        - **System Confidence (SC):** High, Medium, Low (based on data quality, SNN agreement, novelty of the situation).
    - Symbolically, `AffectiveState(t) = {OR(t), DS(t), SF(t), IP(t), SC(t)} = G(SNN_State(t))`, where `G` is the interpretation function.
5. **User Interface & Visualization:**
    - A dynamic dashboard displaying the current Affective State, key contributing factors, and geographical context.
    - An interactive map showing fused data layers and areas of concern highlighted by NAFS.
    - A query interface (inspired by the "AOI Shell") allowing users to ask specific questions or request detailed data for particular regions or parameters. For instance:
        - `query risk_level region_A`
        - `show contributing_factors for OR=High`
        - `project DS next_3_hours if rainfall_pattern_B persists` (more advanced, predictive capability).

Page 3

3. Unconventional Computing Method: Neuromorphic Approach & Rationale

The choice of neuromorphic computing, specifically Spiking Neural Networks (SNNs), is central to NAFS's innovation and its potential to overcome limitations of conventional systems in rapid flood analysis.

A. Rationale for Neuromorphic Computing

1. **Event-Driven Processing for Real-Time Data:** Flood data, especially from sensors and rapidly updating satellite feeds, is inherently event-based. Changes in water levels, new rainfall, or detected inundation are "events." SNNs process information only when new input spikes arrive, making them highly efficient for sparse, temporally dynamic data, unlike traditional ANNs that often process dense data frames continuously. This aligns perfectly with the need for rapid updates.
2. **Low Power Consumption:** Neuromorphic hardware is designed for significantly lower power consumption compared to conventional CPUs/GPUs performing similar tasks. This is

crucial for potential deployment in edge devices or in scenarios where power is limited, and for sustainable large-scale monitoring.

3. **Temporal Information Processing:** SNNs naturally encode and process information in the precise timing of spikes. This allows NAFS to learn and react to the temporal dynamics of flood development (e.g., the rate of water rise, the sequence of events leading to a flash flood) in a more direct way than frame-based systems.
4. **Robustness to Noise and Incomplete Data:** Biological neural systems are remarkably robust. SNNs, inspired by them, have the potential to better handle noisy sensor data or missing information from certain sources, which is common in real-world disaster scenarios. The "System Confidence" output of NAFS would reflect this.
5. **Complex Pattern Recognition and Learning:** SNNs can learn complex, non-linear relationships in high-dimensional spatio-temporal data. This is vital for identifying subtle precursor patterns of severe flooding that might be missed by traditional rule-based systems or simpler machine learning models.

B. Conceptual SNN Operation within NAFS

(Conceptual Diagram 2: Simplified SNN Neuron and Spike Train)

```
Input Spike Trains (from fused data features)
  x1(t) |||| | || ||| |
  x2(t) | || |||| | | |
  x3(t) || | | |||||| |
         \  |  /
          \ | /
  +-------------------+
  |   Leaky Integrate-|--> Membrane Potential V(t) --| Threshold Θ |-->
Output Spike y(t)
  |   and-Fire Neuron |                              |             |      (to
next layer
  |   (LIF Model)     |                              +-------------+       or
Affective State)
  |   w1, w2, w3      |
  +-------------------+
           ^
           | Learning Rule (e.g., STDP) modifies weights w_i
```

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1. **Spike Encoding:** Continuous data from the Fusion Module (e.g., water level, precipitation rate) is converted into spike trains. For example:

- Rate encoding: Higher water level = higher spike frequency.
- Latency encoding: Faster change = earlier spike.
- Threshold-based encoding: Spike generated when a value crosses a critical threshold.
  `X_fused(loc, t) → SpikeTrain_loc(t)`

2. **Synaptic Processing:** Each spike arriving at a neuron `j` from a presynaptic neuron `i` contributes to neuron `j`'s membrane potential, scaled by the synaptic weight `w_ij`. `ΔV_j(t) = Σ_i w_ij * S_i(t-δ_ij)` where `S_i` is a spike from neuron `i` and `δ_ij` is propagation delay.
3. **Neuronal Integration and Firing:** Neurons (e.g., Leaky Integrate-and-Fire model) integrate these weighted inputs over time. If the membrane potential `V_j(t)` exceeds a threshold `Θ_j`, the neuron fires an output spike and its potential is reset.
   `If V_j(t) >= Θ_j, then output_spike_j(t) = 1; V_j(t) = V_reset.`
4. **Learning (e.g., STDP):** Synaptic weights `w_ij` are modified based on the relative timing of pre- and post-synaptic spikes.
   - If pre-synaptic spike slightly precedes post-synaptic spike: `Δw_ij > 0` (Long-Term Potentiation).
   - If post-synaptic spike slightly precedes pre-synaptic spike: `Δw_ij < 0` (Long-Term Depression).
     This allows the network to learn causal relationships and temporal patterns in the input data that signify different flood characteristics.
5. **Population Coding for Affective States:** The final "affective state" outputs (Overall Risk, Dynamic Stability, etc.) would likely be represented by the collective activity of specific neural populations in the SNN. For example, a high, synchronized firing rate in the "Severe Risk" population indicates a critical situation.
   `OverallRisk(t) = decode(Activity(Population_Risk(t)))`

C. Hybrid Considerations with In-Memory Computing

For the initial data-heavy stages (ingestion, pre-processing, basic feature extraction from large satellite images), In-Memory Computing could offer significant speed-ups. By performing computations directly where data is stored (e.g., in specialized DRAM or ReRAM crossbar arrays), it minimizes data movement to a CPU/GPU, reducing latency and energy. This pre-processed, feature-rich data can then be more efficiently converted to spike trains for the neuromorphic core. This creates a powerful pipeline: massive raw data efficiently crunched by in-memory systems, then nuanced temporal patterns discerned by neuromorphic systems.

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4. Application, Impact, and Alignment with Judging Criteria

A. Application for Target Audience (Emergency Responders)

Emergency responders would interact with NAFS through a dedicated dashboard or a more flexible query interface.

1. **At-a-Glance Situational Awareness:** The "Affective State" provides an immediate, high-level understanding:
    - Instead of sifting through multiple maps and data tables, a responder sees: *"Overall Risk: High; Dynamic Stability: Rapidly Escalating in Sector Gamma; System Confidence: Medium."*
    - This triggers immediate attention to Sector Gamma.
2. **Prioritization of Resources:** NAFS can highlight areas where "Impact Potential" (on population or critical infrastructure) is highest, guiding resource allocation (evacuation teams, medical aid, temporary barriers).
3. **Early Warning & Anomaly Detection:** By learning subtle precursor patterns, the SNN core could flag developing situations *before* they become obvious in traditional data views, improving lead times for warnings. The "Dynamic Stability" metric is key here.
4. **Drill-Down Capability:** If NAFS indicates "System Confidence: Low" for a particular assessment, users can query the system (e.g., `show conflicting_data_sources for sector_gamma_risk`) to understand uncertainties and potentially task further data collection (e.g., drone deployment).

(Conceptual Diagram 3: User Interaction Workflow)

```
+-------------------+    +----------------------+    +-----------------------
-+
| Diverse Real-Time | -> | NAFS                 | -> | "Affective State"
|
| Flood Data Inputs |    | (Neuromorphic Core)  |    | Displayed on Dashboard
|
+-------------------+    +----------------------+    | (Risk, Stability etc.)
|
                                                      +----------------------
-+
                                                                 | (User
notices "High Risk")
                                                                 v
+-----------------------------------+    +-----------------------------------
--+
| User Queries NAFS via Interface:  | -> | NAFS Provides Detailed Context:
|
```

```
| "details region_X impact_type_Y"  |     | (Key contributing sensor readings,
|
+-----------------------------------+     |  predicted water path, confidence)
|
                                          +-----------------------------------
--+
                                                          |
                                                          v
                                          +-----------------------------------
--+
                                          | Informed Decision & Action
|
                                          | (Resource deployment, evacuation)
|
                                          +-----------------------------------
--+
```

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B. Potential Impact

- **Reduced Decision Latency:** Translating complex data into intuitive summaries can shave critical minutes or hours off response times.
- **Improved Resource Allocation:** More targeted deployment of limited resources based on nuanced risk assessment.
- **Enhanced Community Resilience:** Better early warnings and more effective responses can save lives, reduce property damage, and aid faster recovery.
- **Democratization of Advanced Analysis:** NAFS aims to make sophisticated flood analysis accessible beyond highly specialized GIS/hydrology experts.
- **Driving Unconventional Computing Adoption:** Demonstrating a tangible, high-impact application in Earth science can accelerate research and development in neuromorphic and related fields.

C. Alignment with Judging Criteria

1. **Novelty & Innovation:**
    - The core novelty lies in the *translation of complex flood data into "affective" or intuitive risk indicators* using neuromorphic computing. This moves beyond purely quantitative outputs.

- The application of SNNs for holistic, dynamic flood risk assessment, considering temporal patterns and data-source confidence, is a novel approach in this domain.

2. **Feasibility & Technical Merit:**
    - The concept builds upon established principles of neuromorphic computing (SNNs, LIF models, STDP-like learning) and existing research in data fusion.
    - Simulators for neuromorphic hardware (e.g., SpiNNaker, Brian, NEST) and cloud access to early neuromorphic platforms allow for initial prototyping and validation.
    - The phased approach (simulation, then potential hardware implementation) is feasible.
3. **Potential for Impact on Flood Analysis:**
    - Significant potential to improve the speed, clarity, and actionability of flood information for disaster responders, as outlined above.
    - Addresses NASA's goal of "more actionable information at the fingertips of decision-makers."
4. **Clarity and Comprehensiveness of Submission:**
    - This white paper aims to clearly articulate the problem, proposed solution, unconventional computing method, and potential impact.
5. **Utilization of Unconventional Computing:**
    - Neuromorphic computing is central to the proposed solution, addressing its strengths in event-based processing, temporal pattern recognition, and efficiency. In-Memory computing is identified as a complementary technology for data-intensive pre-processing.

Page 5

5. Future Work, Challenges, and Conclusion

A. Proposed Development Phases & Future Work

1. **Phase 1 (6-12 months): Simulation & Algorithm Development**
    - **Dataset Curation:** Assemble or simulate diverse flood datasets (historical and synthetic, incorporating data from NASA resources like FLOODS).
    - **SNN Architecture Design:** Develop and refine the SNN architecture for NAFS in a neuromorphic simulator (e.g., Brian2, Nengo, or using frameworks targeting Loihi-like systems).
    - **Spike Encoding Schemes:** Experiment with different methods for encoding fused flood data into spike trains.
    - **Learning Algorithm Implementation:** Implement and test STDP-based and other relevant learning rules. Train initial models on historical data.

    - **"Affective State" Mapping:** Define and refine the rules and neural population coding for translating SNN activity into the defined affective output parameters.
    - **Proof-of-Concept Dashboard:** Develop a basic interface to visualize simulated NAFS outputs.
2. **Phase 2 (12-24 months): Prototyping on Neuromorphic Hardware & Validation**
    - **Hardware Adaptation:** Port the SNN model to available neuromorphic hardware (e.g., research chips, cloud platforms).
    - **Real-time Data Integration (Simulated):** Develop interfaces to feed simulated real-time data streams into the hardware prototype.
    - **Performance Benchmarking:** Evaluate speed, power consumption, and accuracy against conventional methods using benchmark scenarios.
    - **Validation with Experts:** Engage emergency managers and hydrologists to evaluate the intuitiveness and utility of NAFS outputs. Iterate based on feedback.
3. **Phase 3 (Beyond 24 months): Pilot Deployment & Refinement**
    - **Integration with Live Data Feeds:** If feasible, pilot NAFS with select live (or near-live) data sources in a controlled environment.
    - **Scalability and Robustness Testing:** Assess the system's ability to handle increasing data volumes and diverse scenarios.
    - **Investigate In-Memory Computing Integration:** Develop and test prototypes for in-memory pre-processing modules if initial data bottlenecks are identified.

B. Potential Challenges

- **Neuromorphic Algorithm Maturity:** While promising, developing complex, large-scale SNN applications is still an active research area. Training SNNs effectively can be challenging.
- **Data Availability and Quality:** Access to comprehensive, high-quality, labeled historical flood data for training is crucial.
- **Computational Resources:** Access to neuromorphic hardware or sufficient simulation capacity will be necessary.
- **Validation of "Affective" Outputs:** Quantitatively validating the "intuitiveness" and "correctness" of the affective states will require careful methodology and expert involvement.
- **Scalability:** Scaling the SNN to handle national or global level data while maintaining real-time performance.

C. Conclusion

The Neuromorphic Affective Flood Sentinel (NAFS) offers a transformative approach to flood analysis by harnessing the unique capabilities of neuromorphic computing. By translating complex, multi-modal data into rapidly understandable, "affective" risk assessments, NAFS aims to significantly enhance the ability of emergency responders to make timely and effective decisions. This project directly aligns with NASA's vision of leveraging unconventional computing to solve

critical Earth science problems. While challenges exist, the potential benefits—saving lives, protecting property, and advancing the frontier of applied unconventional computing—make NAFS a compelling endeavor. We believe this concept, rooted in the innovative spirit of systems like the AOI_draw_v4_shell, can mature into a powerful tool for a more resilient future.

D. References (Illustrative - to be populated as needed)

- NASA Earth Science to Action Strategy.
- Relevant publications on neuromorphic computing (e.g., Mead, C., "Neuromorphic Electronic Systems").
- Papers on SNNs for spatio-temporal pattern recognition or sensor fusion.
- Information on NASA FLOODS dataset or the IEEE challenge mentioned.
- Resources on target neuromorphic platforms (Loihi, SpiNNaker, etc.).

1. Proposed Solution Details

A. Explanation of proposed unconventional or hybrid computing solution

I. Detail problem statement for your solution

II. Define technical details and feasibility of the solution

B. Discussion of integration of flood data sets into mapping or modeling solution

I. Specific capabilities, datasets, or tools leveraged

II. How capabilities are applied to future use cases to improve flood preparedness or response?

III. How are the solution methodology and data tailored to flood analysis evaluation or related Earth Science inquiries?

2. Potential Impact

A. Benefits and impact of proposed solution

I. How does this solution build upon current research and literary review

II. What effect does the solution have on speed, accuracy, and other technical variables compared to traditional methods?

III. Scalability and implementation feasibility

IV. Concrete examples and scenarios where the solution outperforms current methods

3. Team Experience and Market Evaluation

A. Qualifications and expertise of team members

B. Past projects or research relevant to proposed solution

C. Analysis for how proposed solution can eventually be scaled or integrated into future markets

I. Identification of potential product users

II. How have criteria from your target user community informed solution design?

III. Analysis of the potential commercial market qualities, landscape, and competing technologies

4. Expected Outcomes

A. Next steps if selected as semi-finalist

I. Goals for prototype development, customer discovery, further research, or other advancement of your technology

5. Conclusion

A. Key points about the proposed flooding technology solution

B. Summary of the proposed solution's strengths and potential impact

6. References (Place on separate page, not counted for 5-page limit)